

Vincent Cavez

Curriculum Vitae

Research Summary

I am a Human-Computer Interaction researcher specializing in the design and study of interactive systems that enrich human activities, from creative practices to learning, while keeping interactions simple and fluid. My work draws on gesture, pen, and voice interaction, and I have a growing interest in theoretical models of interaction that capture how digital tools mediate and shape user activity. I have built systems for music score writing, spreadsheet interaction on interactive surfaces, children's storytelling, spelling instruction, and physics learning through sketching. I recently completed a Postdoctoral Scholarship at Stanford University, where I worked with Hari Subramonyam. I received my Ph.D. in Computer Science from Paris-Saclay University, advised by Emmanuel Pietriga and Caroline Appert. I am exploring opportunities for my next research position.

Research Experience

- 2025–2026 **Postdoctoral Scholar**, *Stanford University, Graduate School of Education*, Stanford, USA.
Host: Hari Subramonyam
Designing AI-driven interactive tools for children's narrative language development through storytelling. Developing pedagogical program synthesis for interactive spelling instruction. Investigating how sketch-based interaction can support physics learning. Exploring theoretical models of interaction that capture how digital tools mediate and shape user activity. Designing tools for asynchronous designer-client collaboration.
- 2020–2025 **Doctoral Researcher**, *Inria Saclay, ILDA Team*, Paris-Saclay University, France.
Advisors: Emmanuel Pietriga and Caroline Appert
Designed and evaluated pen-based interactive systems for productivity and creativity on interactive surfaces, including novel interaction techniques for spreadsheets and music score writing.

Education

- 2025 **PhD**, *Inria Saclay*, Paris-Saclay University, France.
Computer Science (*Human-Computer Interaction*)
Thesis: *Designing Pen-based Interactions for Productivity and Creativity*
Advisors: Emmanuel Pietriga and Caroline Appert
<https://theses.hal.science/tel-04986694>
- 2020 **Master's Degree**, Paris-Cité University, France.
Mathematical Engineering and Statistics - with high honours
Thesis: *Contributions to the development of an embedded object finder for the visually impaired*
Advisors: Antonin Goude and Oscar De La Cruz
- 2019 **University Degree**, Paris-Saclay University, France.
Creation and Development of Innovative Start-ups
- 2018 **Bachelor's Degree**, Paris-Saclay University, France.
Mathematics - Modelization Specialization

Publications

In Preparation & Under Review

- 2026 **Vincent Cavez**, Marielle Zheng, Momin Siddiqui, Abbie L Olszewski, Srirangaraj Setlur and
UIST Hari Subramonyam. 2026.
Tellimations: Just-in-time Scene Element Animation for Supporting Children's Story-telling. *Under review.*
- 2026 Xiaohan Peng and **Vincent Cavez**. 2026.
UIST **Cross-expertise Remote Early Prototyping Exchange.** *Under review.*

Peer-Reviewed

- 2026 **Vincent Cavez**, Kashif Imteyaz and Anne-Flore Cabouat. 2026.
DIS EA **Structural Interaction: Shifting the Focus of User Interface Design.** DIS Companion
'26, Designing Interactive Systems Conference, June 13–17, 2026, Singapore.
<https://doi.org/10.1145/3802974.3809476>
- 2026 Xintian Fu, **Vincent Cavez**. 2026.
CHI EA **Opportunities to Support Musicians' Score-based Practice with Context-Specific
Annotations on Tablet.** Extended Abstracts of the 2026 CHI Conference on Human Factors
in Computing Systems (CHI EA '26), April 13–17, 2026, Barcelona, Spain.
<https://dl.acm.org/doi/10.1145/3772363.3798332>
- 2026 Momin N. Siddiqui, **Vincent Cavez**, Sahana Rangasrinivasan, Abbie L Olszewski, Srirangaraj
IUI Setlur, Maneesh Agrawala and Hari Subramonyam. 2026.
**Teaching Spell Checkers to Teach: Pedagogical Program Synthesis for Interactive
Learning.** To appear in IUI '26: Proceedings of the 31st International Conference on Intelligent
User Interfaces.
<https://arxiv.org/abs/2512.12115>
- 2025 **Vincent Cavez**, Catherine Letondal, Caroline Appert and Emmanuel Pietriga. 2025.
CHI **EuterPen: Unleashing Creative Expression in Music Score Writing.** In CHI Conference
on Human Factors in Computing Systems (CHI '25), April 26–May 01, 2025, Yokohama,
Japan. ACM, New York, NY, USA, 16 pages.
<https://doi.org/10.1145/3706598.3713488> (acceptance rate: 24.9%, 1249/5014)
- 2024 **Vincent Cavez**, Catherine Letondal, Emmanuel Pietriga, and Caroline Appert. 2024.
CHI **Challenges of Music Score Writing and the Potentials of Interactive Surfaces.** In
Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems (CHI '24).
Association for Computing Machinery, New York, NY, USA, Article 728, 1–16.
<https://doi.org/10.1145/3613904.3642079> (acceptance rate: 26.3%, 1060/4028)
- 2024 **Vincent Cavez**, Caroline Appert, and Emmanuel Pietriga. 2024. **Spreadsheets on Interac-
TOCHI tive Surfaces: Breaking through the Grid with the Pen.** ACM Trans. Comput.-Hum.
Interact. 31, 2, Article 16 (April 2024), 33 pages.
<https://doi.org/10.1145/3630097>

Workshop Papers

- 2026 **Vincent Cavez**. 2026.
CHI WS **Structural Interaction for Generative User Interfaces.** In *Generative UI meets HCI*,
Workshop at the CHI Conference on Human Factors in Computing Systems (CHI '26), April
13–17, 2026, Barcelona, Spain. CEUR Workshop Proceedings.
https://www.vincentcavez.com/pdf/Structural_Interaction_Gen_UI.pdf

2026 **Vincent Cavez**. 2026.
CHI WS **How Structure Mediates Creative Activity Traces**. In *Herding CATs: Making Sense of Creative Activity Traces*, Workshop at the CHI Conference on Human Factors in Computing Systems (CHI '26), April 13–17, 2026, Barcelona, Spain.
https://www.vincentcavez.com/pdf/Structural_Interaction_CAT.pdf

Posters

2026 **Teaching Spell Checkers to Teach: Pedagogical Program Synthesis for Interactive Learning**. LSTD@25, Stanford Graduate School of Education
<https://www.vincentcavez.com/pdf/LSTDPoster.pdf>

2024 **Designing Pen-based Interactions for Productivity and Creativity**. RJC IHM 2024, RJC IHM AFIHM
<https://www.vincentcavez.com/pdf/RJCPoster.pdf>

Mentoring

03/2026–**Xintian Fu**, *MSc, HCI*, Paris-Saclay University, France.
present Interacting with Physical Objects in Mixed Reality via Gaze Snapshots
10/2025–**Marielle Zheng**, *MSc, Learning Design and Technology*, Stanford University, USA.
03/2026 AI-assisted storytelling for children with developmental language disorder
09/2025–**Anika Rajaram**, *High school intern, Harker School*, Stanford University, USA.
present Learning mechanics through sketch-generated widgets
06/2025–**Momin Siddiqui**, *Visiting RA, Georgia Tech*, Stanford University, USA.
01/2026 Spelling support for children with dyslexia via structured word inquiry
06/2025–**Kashif Imteyaz**, *Visiting PhD, Northeastern University*, Stanford University, USA.
08/2025 Sketch-and-code approach for physics game development
05/2025–**Xintian Fu**, *MSc, HCI*, Paris-Saclay University, France.
08/2025 Context-specific annotations for music score practice on tablet
02/2024–**Noémie Hanus**, *MSc, HCI*, Paris-Saclay University, France.
08/2024 Itinerary-driven map navigation on mobile

Teaching

2024–2025 **Teaching Assistant - Design of Interactive Systems, M1–M2 HCI**, Paris-Saclay University.
Professor: Wendy Mackay
2024–2025 **Teaching Assistant - Introduction to HCI, L2 CS**, Paris-Saclay University.
Professor: Michel Beaudouin-Lafon
2022–2024 **Teaching Assistant - Winter School, M1 HCI**, Paris-Saclay University.
Professor: Vanessa Peña-Araya
2015–2017 **Teaching Assistant - Mathematics, 7th to 9th Grade**, Mondétour Middle School.
Teacher: Marie Decamps

Outreach

04/2026 **PreCHI Paris**, *Organizer, Session Chair*, Saclay, France.
<https://sites.google.com/view/paris-pre-chi26/home>
02/2026 **Learning Sciences and Technology Design 25th Anniversary Conference, Poster**, Stanford, USA.
<https://ed.stanford.edu/events/lstd-25>
11/2025 **ASHA Convention**, *Invited talk*, Washington DC, USA.
<https://convention.asha.org/>

- 04/2025 **ParaCHI Paris**, *Organizer, Speaker, Session Chair*, Paris, France.
<https://sites.google.com/view/para-chi-paris25>
- 02/2025 **VISVAR Research Group**, *Invited talk*, Stuttgart, Germany.
- 10/2024 **11th Young Researchers in Human-Computer Interaction Meeting**, *Poster*, Saint-Pierre-d'Oléron, France.
<https://rjc2024.afihm.org/>
- 05/2024 **Foundation Interface Lab**, *Invited talk*, San Diego, USA.
- 02/2024 **Meetup of Junior Women in Maths and Computer Science**, *Animator*, Saclay, France.
<https://filles-et-maths.fr/rjmi/>
- 12/2023 **2024 3MT® Paris-Saclay University Competition**, *Finalist*, Saclay, France.
<https://www.universite-paris-saclay.fr/three-minute-thesis-competition-3mtr#home200832>
- 10/2023 **PreVis '23**, *Demo*, Saclay, France.
<https://www.aviz.fr/Events/Previs2023>
- 12/2022 **LISN's HCI Department Day**, *Speaker*, Saclay, France.
- 06/2022 **LISN Ph.D. Days**, *Organizer, Speaker*, Saclay, France.

Service

Reviewing

- 2026 **C&C 2026**, *Reviewer*.
- 2026 **CHI 2026**, *Reviewer, Associate Chair*.
- 2025 **VIS 2025**, *Reviewer*.
- 2025 **CHI 2025**, *Reviewer*.
- 2024 **CHI 2024**, *Reviewer, Session Chair*.

Elected Roles

- 2024–present **ACM SIGCHI Paris Chapter**, *Elected Webmaster*.
- 2024–2025 **Paris-Saclay University Council of the Doctoral School**, *Elected Representative*.
- 2024–2025 **Paris-Saclay University Pole Committee of the Doctoral School**, *Elected Representative*.
- 2024–2025 **Inria Saclay Center Committee**, *Elected Representative*.

Funding

- 2025–2026 Stanford HAI Postdoctoral Fellowship. PI: Hari Subramonyam. \$80,000.
- 2021–2024 Inria CORDI-S Doctoral Fellowship.

Awards & Honors

- 2026 **2 Special Recognition for Outstanding Reviews**, *CHI 2026*.
- 2024 **Special Recognition for Outstanding Review**, *CHI 2024*.

Extra-Curricular

- Since 2010 Music composition: electronic and orchestral music, live performance, film soundtracks.
<https://lordaaron.bandcamp.com/>

Technical Skills

- Programming JavaScript (D3.js, React), Python (OpenCV, TensorFlow), R (Tidyverse), Swift, Go
- Design & Media Figma, Logic Pro X, Ableton Live, Adobe Premiere Pro, Final Cut Pro, Blender

Languages French (native), English (fluent)

Languages

French **Native**

English **Fluent**